

SYED MUTAHIR HUSSAIN

AI & Data Engineer

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PROFESSIONAL SUMMARY Final-year Computer Science student specializing in Machine Learning, NLP, and Data Science. Experienced in building end-to-end ML pipelines, fine-tuning models, and deploying AI solutions for international clients. Strong foundation in Python, SQL, and algorithm design, with a focus on writing clean, production-ready code.

PROFESSIONAL EXPERIENCE

Independent AI Developer | Self-employed | 2023 – Present

- Engineer and deploy custom end-to-end Machine Learning pipelines and Generative AI solutions.
- Developed a high-performance RAG pipeline utilizing **LangChain** and **Pinecone**, optimizing context retrieval through hybrid search.
- Design automated Python-based data extraction and sentiment analysis workflows for complex datasets.
- Manage the full deployment lifecycle, from model selection and hyperparameter tuning to production-ready API integration.

Cinematic Animator Intern | Unity Development | 2024

- Scripted custom animation states and handled scene transitions within the **Unity Engine** using C#.
- Optimized rendering settings to maintain stable framerates across different hardware constraints.

SELECTED PROJECTS Accident Report Narrative Generator (NLP / Python)

- Developed a specialized NLP tool that takes raw numerical data from crash sensors and translates it into clear, readable narrative reports.
- Built an automated export feature that compiles batches of 10 accident cases into neatly formatted PDF files for official record-keeping.

Archimedes RAG Engine (LLMs / LangChain)

- Built a Retrieval-Augmented Generation (RAG) system using LangChain and Pinecone vector databases to chat with proprietary documents.
- Added a custom hybrid search and re-ranking approach, which improved the accuracy of the retrieved information by roughly 40%.

Titan Vision Analytics (Computer Vision / YOLOv10)

- Trained and deployed a YOLOv10 computer vision pipeline for real-time object tracking and safety monitoring.
- Optimized the model to handle 4k/60fps video streams smoothly by utilizing hardware acceleration and CUDA.

Spectre MLOps Monitor (DevOps / Python)

- Set up a monitoring system to track the accuracy of deployed machine learning models over time.
- Used FastAPI and Prometheus to automatically trigger alerts and retrain the model whenever performance dropped below acceptable thresholds.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, SQL, TypeScript
- **AI & Machine Learning:** PyTorch, LangChain, YOLOv10, OpenCV, Scikit-Learn, XGBoost
- **Databases:** PostgreSQL, MySQL, Pinecone, Milvus
- **Tools & Infrastructure:** Docker, FastAPI, Git/GitHub, Linux

EDUCATION **University of Engineering and Technology (UET), Taxila** Bachelor of Science in Computer Science | Expected Graduation: 2026

Current Status: 8th Semester

Core Coursework: Artificial Intelligence, Data Structures & Algorithms, Object-Oriented Programming, Database Systems.